

MATH0039 Differential and Integral Calculus

<i>Year:</i>	2025–2026
<i>Code:</i>	MATH0039
<i>Level:</i>	4 (UG)
<i>Normal student group(s):</i>	UG: Students outside Mathematics
<i>Value:</i>	15 credits (= 7.5 ECTS credits)
<i>Term:</i>	1
<i>Assessment:</i>	85% examination, 15% coursework
<i>Normal Pre-requisites:</i>	A-level Maths or strong GCSE
<i>Lecturer:</i>	Teymour Gray & Elvar Atlason

Course Description and Objectives

This course provides a fairly rapid introduction to calculus. Calculus underlies almost all areas of mathematics and a great deal of science and engineering. The aim of the course will be to provide a solid grounding in this fundamental branch of mathematics for students who have a limited mathematical background.

Most of the course material occurs in A-level mathematics: the treatment here will be slightly more advanced. The course is suitable for students with good GCSE mathematics (or a weak A-level pass).

Recommended Texts

Croft, Davison and Hargreaves, *Introduction to Engineering Mathematics* (Addison-Wesley)

Detailed Syllabus

1. **Functions and graphs**

Polynomials, exponentials and logarithms, trigonometric functions.

2. **Differentiation**

Differentiation from first principles, applications of differentiation, implicit differentiation, Taylor series.

3. **Integration**

Fundamental theorem of calculus, indefinite and definite integrals, calculating area, integration by parts and substitution.

4. **Differential equations**

First order differential equations, separation of variables and integrating factors, second order differential equations, complementary functions and particular integrals, numerical approximations, physical applications.